INCH-POUND
MIL-PRF-49166/3A(CR)
3 June 1999
SUPERSEDING
MIL-S-49166/3(CR)
8 April 1982

# PERFORMANCE SPECIFICATION SHEET
# MECHANICAL SCANNER, MX-9872(V)3/UA

This performance specification sheet is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense. The complete requirements for procuring the MX-9872(V)3/UA Scanner shall consist of this document and the General Specification, MIL-PRF-49166.

## 1. SCOPE

1.1 Scope. This performance specification covers the Mechanical Scanner, MX-9872 (V) 3/UA and augments MIL-PRF-49166. The MX-9872(V) 3/UA is used on the AN/VSG-2, Tank Mounted Thermal Sight.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications. The following specifications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

AMSC N/A FSC 5855

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SPECIFICATIONS

DEPARTMENT OF DEFENSE

| | |
|---|---|
| MIL-PRF-49166 | - Mechanical Scanner, MX-9872(V)/UA; General specification for |
| MIL-PRF-49177 | - Scan-Interlace, PL-1408/UA |

(Unless otherwise indicated, copies of the above specification are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

USA CECOM

| | |
|---|---|
| SM-D-807690-3 | Scanner, Mechanical MX-9872(v)3/UA |
| SM-D-657713 | Cover, Scanner Housing |
| SM-D-657620 | Housing, Scanner |

(Copies of documents, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Where specified in the contract (see 6.2), the contractor shall furnish first article units for inspection (see 4.2).

3.2 Materials. The contractor shall select the materials, but the materials and parts used shall be fully capable of meeting all of the operational and environmental requirements specified herein. Selection criteria of the class, grade or type part shall be that the material will be able to perform its



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intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard.

3.3 Description. The MX-9872(V)3/UA Scanner, herein known as the MX-9872(V)3/UA, shall consist of the MIL-PRF-49166, MX-9872(V)/UA Scanner with a phase-shift lens and counter weight assembled and adjusted for 42 Hz operation.

3.4 Construction. The MX-9872(V)3/UA shall be constructed in accordance with drawing SM-D-807690-3, Mechanical Scanner Assembly, and shall meet the form, fit, function, and interface requirements of this specification.

3.4.1 Weight. Weight of the MX-9872(V)3/UA shall be 1.55 pounds maximum when weighed with a scale having an accuracy of 1% or better.

3.4.2 Dimensions. The dimensions and dimensional tolerances of the MX-9872(V)3/UA shall be in accordance with Drawings SM-D-807690-3 (Mechanical Scanner Assembly), SM-D-657713 (Cover, Scanner Housing) and SM-D-657620 (Housing Scanner) to ensure items built IAW this specification physically interface with existing hardware.

3.4.3 Components.

a. Scanner, Mechanical MX-9872(V)/UA

b. Phase-shift lens assembly

c. Counterweight

3.5 Performance characteristics. The characteristics specified herein shall apply as defined in 3.1 through 3.5 described herein except that the particular tests required in MIL-PRF-49166, including modular transfer function (4.5.8), shock (4.6.4) and vibration (4.6.5) will apply with the phase shift lens, counterweight and associated hardware removed. Operation shall be at 42 Hz using the Scan-Interlace, PL-1408/UA.

3.5.1 Scan angle and frequency.

3.5.1.1 Variable scan frequency. Not applicable.

3.5.1.2 Specific scan frequency (42 Hz). Scan frequency of the MX-9872(V)3/UA shall be adjusted to 42.0, ±0.5 Hz. Scan frequency drift shall not be greater than ±4.2 Hz. Scan frequency drift shall not be greater than ±4.2 Hz from the frequency at ambient setting (see MIL-PRF-49166 para. 3.14.3) when exposed to high and low temperature environmental conditions (see MIL-PRF-49166).

3.5.1.3 Synchronous scan frequency. Not applicable.



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3.5.1.4 Scan angle. The scan angle shall be 12.8, ±0.5 degrees.

3.5.2 Interlace. The MX-9872(V)3/UA shall operate in the 2:1 interlace mode. The interlace switching time shown in MIL-PRF-49166, Figure 1, shall not be greater than 3.6 milliseconds (ms).

3.5.3 Scan direction. The MX-9872(V)3/UA shall show the waveform shapes and phase relationships in Figure 1 while operating in the 2:1 rear interlace mode.

3.5.4 Power consumption. Power required to operate the MX-9872(V)3/UA shall not exceed 7.5 watts.

3.5.5 Scan efficiency. The MX-9872(V)3/UA scan efficiency shall be a minimum of 65 percent.

3.5.6 Scan jitter in azimuth. Jitter time shall be 18.0 microseconds maximum.

3.6 Environmental conditions. See MIL-PRF-49166 except as follows:

3.6.1 Vibration. The MX-9872(V)3/UA shall not be damaged by vibration over a frequency spectrum at the specified g levels and amplitudes shown in Figure 2 for the X axis and as shown in Figure 3 of MIL-PRF-49166 for the Y and Z axes. During specified vibration, the average interlace waveform amplitude shown in MIL-PRF-49166, Figure 1 shall not deviate from the amplitude without vibration by more than ±50 percent.

3.7 Workmanship. All electronic parts, components, assemblies or sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires and integrated circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting. MIL-HDBK-454 may be used as guidance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)

b. Conformance inspection (see 4.3)

4.2 First article inspection. First article inspection shall be performed on a specified number of scanners when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests of 4.5.1.2 and 4.5.3 included herein; and the test of MIL-PRF-49166 herein referred to in paragraphs 4.5.1.4, 4.5.2, 4.5.4, 4.5.5, 4.5.6, and 4.6 referenced herein.



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4.3 Conformance inspection.

4.3.1 Tests. Perform the examination of 4.4 and the tests of 4.5 et el.

4.4 Examination. Each mechanical scanner shall be examined for compliance with the requirements specified in 3.2 through 3.5. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Methods of inspection. Tests shall be performed utilizing the Scan-Interlace, PL-1408/UA, with the MX9872(V)3/UA with a scan angle of 12.8, ± 0.5 degrees at the 42.0, ± 0.5 Hz scan rate. Otherwise, the contractor/supplier may use his own test methods, measuring devices, tools, fixtures, and other equipment during the conduct of verification testing. This specification cites no test requirements as long as all the performances requirements per Section 3 are fully verified as being met.

4.5.1 Variable scan frequency. Not applicable.

4.5.1.2 Specific scan frequency(42 Hz). The MX-9872(V)3/UA shall be connected to the Scan-Interlace, PL-1408/UA (MIL-PRF-49177), and mounted to a suitable fixture. Power shall be applied and the output of the azimuth (scan) transducer (Scan-Interlace, PL-1408/UA) shall be monitored with a calibrated frequency counter. The output of the Scan-Interlace (PL-1408/UA) shall be adjusted to obtain the specific scan frequency. Failure to meet the requirements of 3.5.1.2 shall constitute failure of this test.

4.5.1.3 Synchronous scan frequency. Not applicable.

4.5.1.4 Scan angle. See MIL-PRF-49166.

4.5.2 Interlace. See MIL-PRF-49166.

4.5.3 Scan direction. With the MX-9872(V)3/UA operating, the outputs of the azimuth and interlace position transducers (Scan-Interlace, PL-1408/UA) shall be observed with an oscilloscope. The output waveforms shall show the shape and phase relationships shown in Figure 1. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Power consumption. See MIL-PRF-49166.

4.5.5 Scan efficiency. See MIL-PRF-49166.

4.5.6 Scan jitter in azimuth. See MIL-PRF-49166.



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4.6 Environmental tests. See MIL-PRF-49166 . Failure to meet the requirements for Vibration as specified by para. 3.6.1 and Figure 2 of this specification, and MIL-PRF-49166B(CR), shall constitute failure of this test..

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's system Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The MX-9872(V)3/UA is intended for use with the AN/VSG-2, Tank Mounted Thermal Sight.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Refer to the General specification (MIL-PRF-49166) for other acquisition requirements data.

6.3 Definitions. (See MIL-PRF-49166)

6.4 Subject term (key word) listing.

Phase-shift lens
Counterweight
Scan-Interlace

6.5 Changes form previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

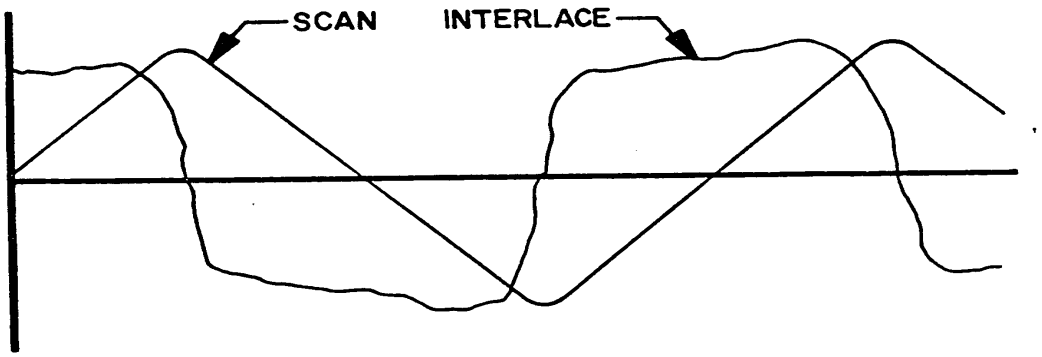
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2:1 REAR POSITION
INTERLACE LEADS SCAN APPROX. 90°




NOTES:
PL-1408/UA- CONNECT P1-33 TO P1-34 AND P1-4 TO P1-17
OSCILLOSCOPE- CONNECT CHANNEL 1 (SCAN POSITION) HIGH INPUT TO J3 (OR J2-7) AND LOW SCOPE TO GND
- CONNECT CHANNEL 2 (SCAN INTERLACE) HIGH INPUT TO J6 (OR J1-9) AND LOW SCOPE TO GND

FIGURE 1. INTERLACE TRANSDUCER AND AZIMUTH POSITION TRANSDUCER OUTPUTS.



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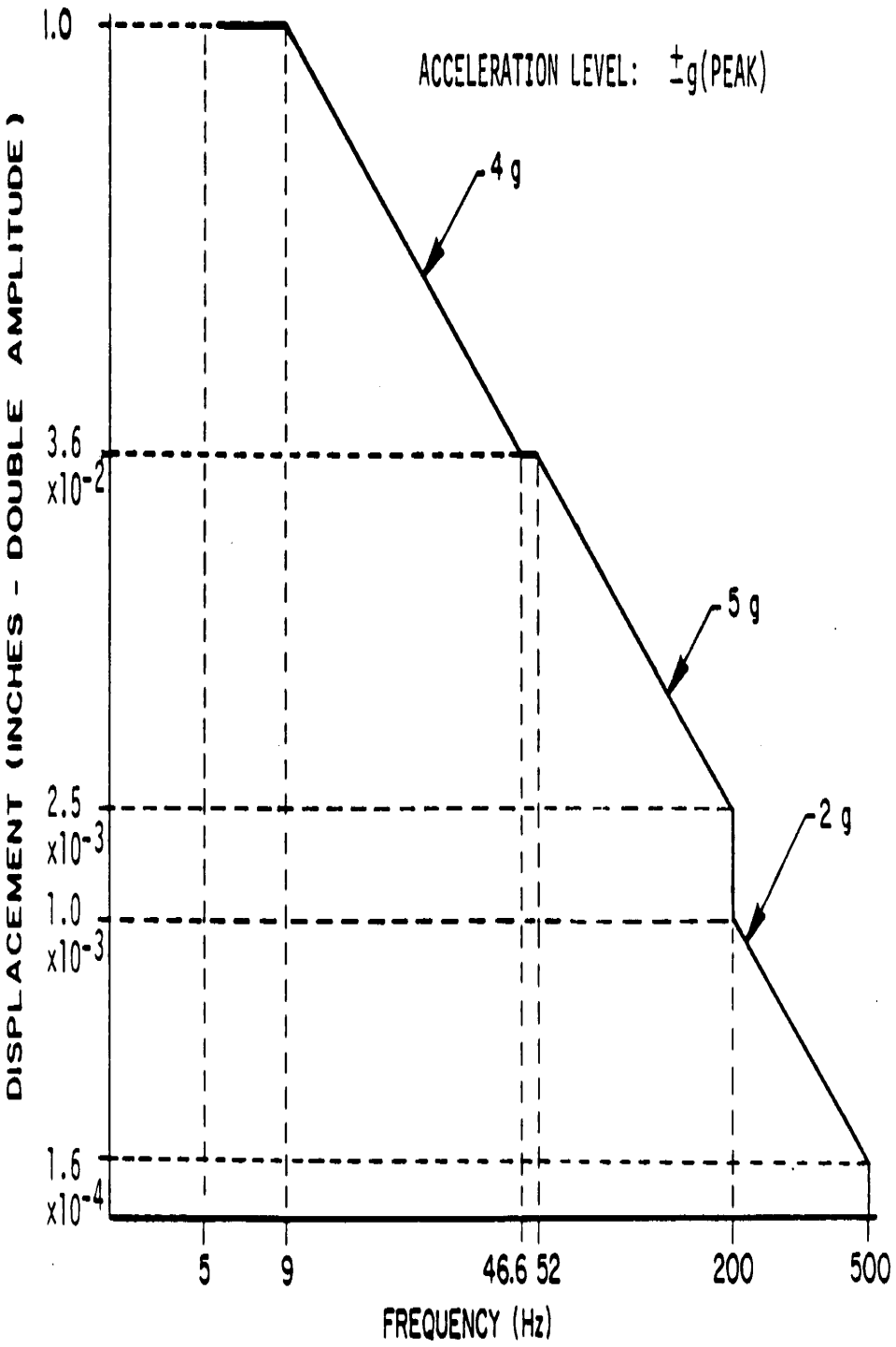



FIGURE 2. VIBRATION TEST PROFILE (X AXIS).



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